

Nicholas Ketz PhD.

Applied Research Scientist: Artificial Intelligence, Bio-informatics, Mathematical Models, Human Studies

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Summary

I am an applied scientist developing machine learning solutions to advanced research problems. I have more than 10 years of experience in both academic and industrial research labs. I have experience individually contributing, leading small teams and coordinated across multiple departments and organizations throughout the entire AI product life cycle.

Relevant Experience

- Large Language Modeling** 03/2022 - Present
 - Developing machine learning solutions for gene therapy and cross-species engineering applications in biotechnology
- Generative AI** 01/2020 - 01/2022
 - Created AI systems for generating video and images grounded in learned human concepts and semantic understanding
- Model Based Reinforcement Learning** 01/2020 - 01/2022
 - Developed Agent-based domain adaptation techniques within autonomous driving using model-based reinforcement learning
- Lifelong Deep Learning** 09/2018 - 01/2021
 - Researched methods for sequential multitask, multi-modality learning deep neural networks to preserve performance across tasks
- Closed-loop Neural Stimulation** 09/2016 - 06/2018
 - Developed devices and algorithms for closed-loop neural stimulation in human subjects to enhance cognitive performance

Education/Employment

- Principle AI/ML Scientist** *Colossal Biosciences/Form Bio* 02/2022 - Present
 - Leading AI/ML research in cross-discipline team on the full AI product life cycle to support various biotech clients
- Research Scientist: Information and Systems Sciences** *HRL Laboratories* 09/2016 - 01/2022
 - Basic AI research funded by Department of Defense and product development for aerospace and autonomous driving
- PhD: Computational Cognitive Neuroscience** *University Colorado, Boulder* 09/2010 - 09/2016
 - Development of biologically inspired neural network informed by neuro-imaging human experiments of cognitive function
- Research Assistant: Davachi Human Memory Lab** *New York University* 09/2007 - 07/2010
 - Supported graduate level research for neuro-imaging of human long-term memory systems
- BA: Physics, minor: Psychology** *University Minnesota, Twin Cities* 09/2003 - 06/2007
 - Studied physics with a minor in psychology, with research experience in particle physics and cognitive-neuroscience

Skills

- Technical:** Deep Learning (CNN, RNN, VAE, Transformers, Multi-Modal, RL), Machine Learning (Optimization, Probabilistic Inference, Unsupervised Learning), Statistics (Parametric/Non-parametric, Bayesian, Time-series), Programming (Python, PyTorch, TensorFlow, Scikit-learn, Git, Docker, R, CUDA, GCP, MLOps)
- AI Lifecycle:** Full-cycle AI development (proof-of-concept to production), cross-team product development, system design/orchestration with containerization, CI/CD pipelines with unit testing and quality assurance
- Professional:** Team leadership, cross-departmental collaboration (engineering, product, sales), client/stakeholder management, project coordination across multiple research domains and organizations

Select Publications/Patents/White Papers

- System and method for semantically grounded video generation** US Patent, 2024
 - Generative video AI system prompted by human interpretable neuro-symbolic concepts
- Model Interpretability Methods to Predict Gene Therapy Manufacturing Failures** White Paper, 2022
 - Extracting learned motifs from DNA sequence models predictive of gene therapy manufacturing failures
- Independent component selection for automated signal artifact removal** US Patent, 2020
 - Scalable solution for removal of nuisance components in time-series data
- Sliced cramer synaptic consolidation for preserving deeply learned representations** ICLR, 2019
 - Framework for overcoming neural network forgetting and improving transfer in sequential multitask learning